Parishram 2.0 2026 (Class 12th)

Biology (BOTANY)

DPP:04

Sexual reproduction in flowering plants

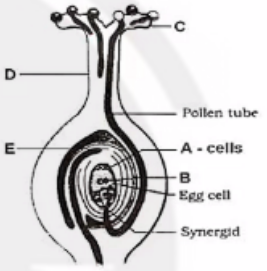
Q1 Which of the following options is correct?
(A) Pollination gives the guarantee of the promotion of post-pollination events that lead to fertilisation.
(B) The events - "from pollen deposition on stigma until pollen tubes enter the ovule" are together referred to as pollen-pistil interaction.
(C) Pollen-pistil interaction is a dynamic process involving pollen recognition followed by only promotion (not rejection) of the pollen.
(D) Pistil has no ability to recognise the pollen, whether right or wrong type.

Q2 The ability of the pistil to recognize the pollen followed by its acceptance or rejection is the result of a continuous dialogue between pollen grain and the pistil. This dialogue is mediated by chemical components of the pollen interacting with those of The pistil. Which of the following chemicals mainly takes part in the interaction:
(A) Nucleotides
(B) Proteins
(C) Minerals
(D) Lipid or inulin

Q3 In flowering plants, process of Self incompatibility can prevent -
(A) Autogamy
(B) Geitonogamy
(C) Xenogamy
(D) Both (a) and (b)

Q4 Inbreeding depression in flowering plants is due to:
(A) continued self-pollination.
(B) cross pollination.
(C) compatible pollens.
(D) hybridisation.

Q5 Identify A, B, C, D and E–



	a	b	c	d	e
A.	Antipodal cells	Secondary nuclei	Stigma	Style	Chalaza
B.	Antipodal cells	Secondary nuclei	Style	Stigma	Chalaza
C.	Antipodal cells	Secondary nuclei	Stigma	Chalaza	Style
D.	Antipodal cells	Secondary nuclei	Chalaza	Stigma	Style

(A) A (B) B
(C) C (D) D

Q6 The events in pollen-pistil interaction are given below
(i) Pollen grain germinates to form pollen tube.
(ii) Pollen tube enters the ovule through micropyle.
(iii) Pollen grain reaches the stigma
(iv) Pollen tube grows through the tissues of stigma and style and reaches the ovary.
The sequential order of their occurrence is

(A) (iii) (iv) (ii) (i)
(B) (ii) (iv) (iii) (i)
(C) (iii) (i) (iv) (ii)
(D) (ii) (iii) (i) (iv)

Q7 Emasculated flowers have to be covered to prevent contamination of its stigma with unwanted pollen. This process is called....
(A) Packing (B) Covering
(C) Removing (D) Bagging

Q8 What is the genetic importance of outcrossing?
(A) Outcrossing is a characteristic of pollinators
(B) Outcrossing increases genetic diversity in a population
(C) Outcrossing increases the chances of sterility
(D) Outcrossing promotes inbreeding in a population

Q9 What serves as the landing platform for pollen grains?

(A) Style (B) Stamen
(C) Ovary (D) Stigma

Q10 What is the role of the breeder in artificial hybridization?
(A) To ensure that only the desired pollen is used for pollination
(B) To carry out natural pollination
(C) To remove both male and female reproductive organs
(D) To prevent fertilization completely

Answer Key

Q1	B
Q2	B
Q3	D
Q4	A
Q5	A
Q6	C
Q7	D
Q8	B
Q9	D
Q10	A

Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

Pollination does not guarantee fertilization; fertilization depends on successful pollen germination and the growth of the pollen tube to the ovule.Pollen-pistil interaction involves both the recognition and the possible rejection of incompatible pollen types, not just promotion.The pistil can recognize whether the pollen is compatible or not, and there are mechanisms in place to reject incompatible pollen.

Video Solution:

Q2 Text Solution:

The ability of the pistil to recognize the pollen followed by its acceptance or rejection is the result of a continuous dialogue between pollen grain and the pistil. This dialogue is mediated by chemical components of the pollen interacting with those of the pistil. Proteins are components which take part in interaction.

Video Solution:

Q3 Text Solution:

In some other species, the anther and stigma are placed at different positions so that the pollen cannot come in contact with the stigma of the same flower. Both these devices prevent autogamy. The third device to prevent inbreeding is self-incompatibility. This is a genetic mechanism and prevents self-pollen (from the same flower or other flowers of the same plant) from fertilising the ovules by inhibiting pollen germination or pollen tube growth in the pistil.

Video Solution:

Q4 Text Solution:

Inbreeding depression in flowering plants is due to continued self-pollination.

Video Solution:

Q5 Text Solution:

1. Antipodal cells – Three cells at the chalazal end of the embryo sac; part of the female gametophyte.
2. Secondary nuclei – Fusion product of two polar nuclei in the central cell; forms endosperm after triple fusion.
3. Stigma – Receptive tip of the pistil where pollen grains land and germinate.
4. Style – Slender stalk connecting stigma to ovary; guides pollen tube to ovule.
5. Chalaza – Basal region of the ovule opposite the micropyle; gives rise to antipodal cells and supplies nutrition.

Video Solution:



Q6 Text Solution:

(iii) (i) (iv) (ii)

Video Solution:



Q7 Text Solution:

Emasculated flowers have to be covered with a bag of suitable size, generally made up of butterpaper, to prevent contamination of its stigma with unwanted pollen. This process is called bagging.

Video Solution:



Q8 Text Solution:

Outcrossing, which is the process of cross-pollination or mating between genetically different individuals, plays a crucial role in increasing genetic diversity. This diversity enhances the adaptability and survival of a population by introducing new genetic combinations.

Video Solution:



Q9 Text Solution:

Stigma is a platform were pollen lands.

Video Solution:



Q10 Text Solution:

In such crossing experiments it is important to make sure that only the desired pollen grains are used for pollination and the stigma is protected from contamination (from unwanted pollen).

Video Solution:

